

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
```

```
df=pd.read_csv("customer_retail csv file.csv")
```

```
print(df.head())
```

	InvoiceNo	StockCode	...	CustomerID	Country
0	536365	85123A	...	17850.0	United Kingdom
1	536365	71053	...	17850.0	United Kingdom
2	536365	84406B	...	17850.0	United Kingdom
3	536365	84029G	...	17850.0	United Kingdom
4	536365	84029E	...	17850.0	United Kingdom

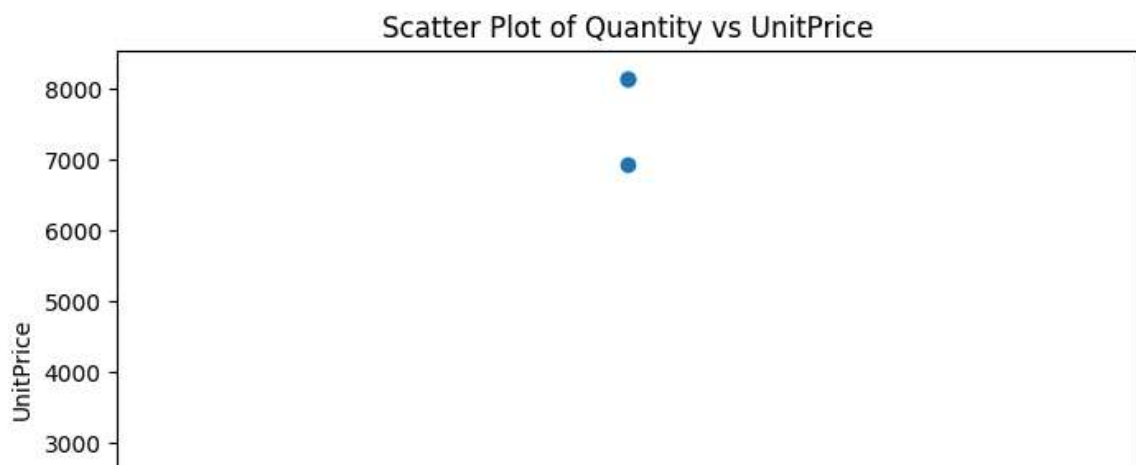
```
[5 rows x 8 columns]
```

```
df=df.dropna()
```

```
df=df[['Quantity','UnitPrice','Country']]
```

```
encoder=LabelEncoder()
df['Country_encoded']=encoder.fit_transform(df['Country'])
```

```
plt.figure(figsize=(8,5))
plt.scatter(df['Quantity'],df['UnitPrice'])
plt.xlabel('Quantity')
plt.ylabel('UnitPrice')
plt.title('Scatter Plot of Quantity vs UnitPrice')
plt.show()
```



```

x=df[['Quantity','UnitPrice']]
y=df['Country_encoded']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_stat

print("Logistic Regression")
log_model =LogisticRegression()
log_model.fit(x_train,y_train)
y_pred=log_model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",accuracy)
cm=confusion_matrix(y_test,y_pred)
print("Confusion Matrix::")
print(cm)

print("Decision Tree")
dt_model=DecisionTreeClassifier()
dt_model.fit(x_train,y_train)
y_pred=dt_model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",accuracy)
print(confusion_matrix(y_test,y_pred))

print("KNN")
knn_model=KNeighborsClassifier()
knn_model.fit(x_train,y_train)
y_pred=knn_model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",accuracy)
print(confusion_matrix(y_test,y_pred))

```

Logistic Regression
 /usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465
 STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regres

```

n_iter_i = _check_optimize_result(
Accuracy: 0.8946152068407672
Confusion Matrix::
[[ 0  0  0 ...  0  70  0]
 [ 0  0  0 ...  0  15  0]
 [ 0  0  0 ...  0   3  0]
 ...
 [ 0  0  0 ...  0   8  0]
 [ 0  0  0 ...  0 23226  0]
 [ 0  0  0 ...  0   13  0]]
Decision Tree
Accuracy: 0.8958477775209922
[[ 6  0  0 ...  0  60  0]
 [ 0  0  0 ...  0  13  0]
 [ 0  0  0 ...  0   3  0]
 ...
 [ 0  0  0 ...  0   8  0]
 [ 20  0  0 ...  0 23163  0]

```

```
[ 0 0 0 ... 0 13 0]]
KNN
Accuracy: 0.8816346968646483
[[ 4 0 0 ... 0 61 0]
 [ 0 0 0 ... 0 13 0]
 [ 0 0 0 ... 0 3 0]
 ...
 [ 0 0 0 ... 0 8 0]
 [ 24 0 0 ... 0 22796 0]
 [ 0 0 0 ... 0 13 0]]
```

```
models=[
    'Logistic Regression',
    'Decision Tree',
    'KNN'
]
accuracies=[
    accuracy,
    accuracy,
    accuracy]
```

```
plt.figure(figsize=(8,5))
plt.bar(models,accuracies)
plt.xlabel('Models')
plt.ylabel('Accuracy')
plt.title('Model Accuracy Comparison')
plt.show()
```

